Futures Time-Series Momentum

ORIE 5260 · Midterm Project Presentation

Data validation, strategy construction, attribution, and risk forecasting

58 futures markets · Data: 1969–2014

Main evaluation: 1976–2014

Based on midterm_report.md, project code, and existing outputs

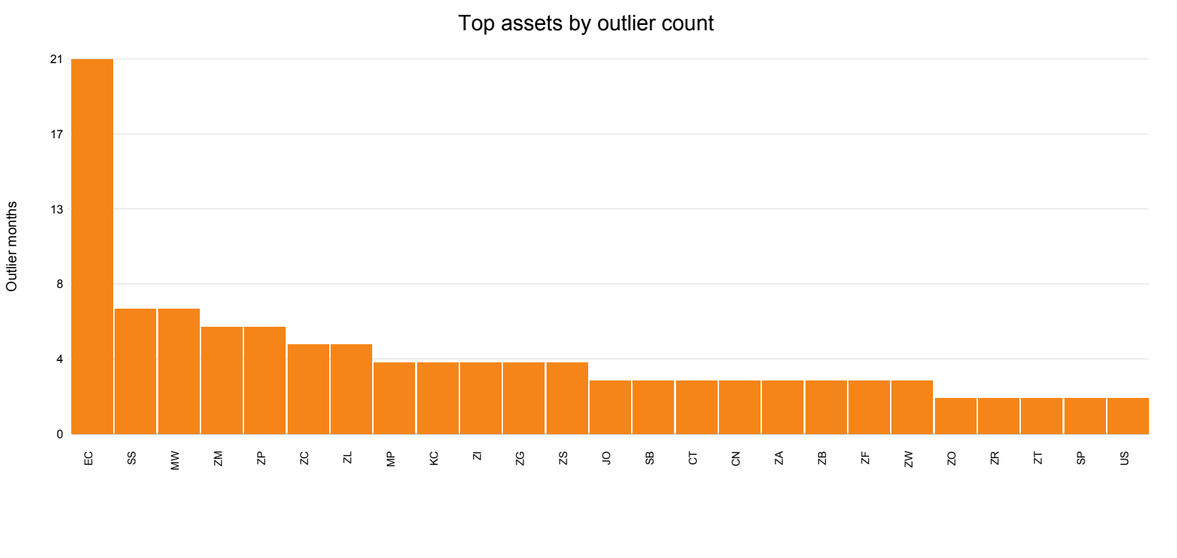
Research question and data

Do past return directions predict next-month futures returns?

Universe	Data
58 markets	26 commodities, 14 equities, 10 fixed income, 8 FX
1969–2014	552 monthly dates; 20,987 valid observations
Unequal histories	551 months at the longest; 75 at the shortest

We evaluate two baselines, then examine returns, risk, and improvements.

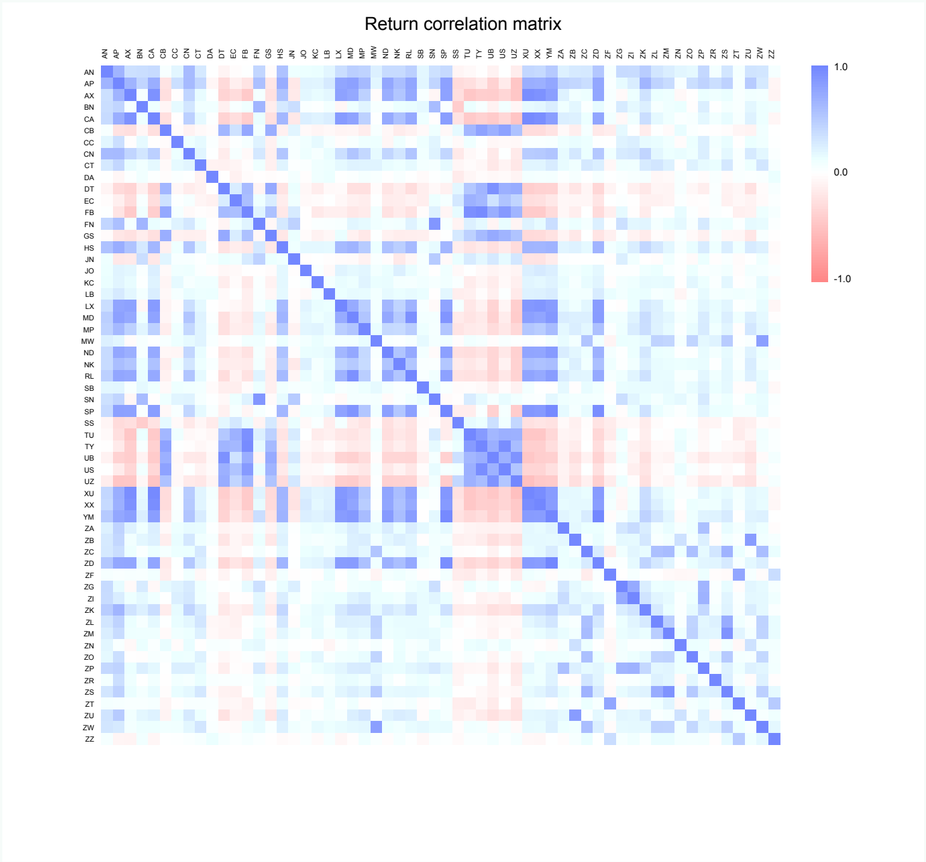
Data quality: flags, reconstruction, and open checks



- 126 outliers flagged with 3×IQR or |robust z| > 5; retained for review.
- 14,974 daily-to-monthly overlapping checks; typical differences approximately zero.
- RL maximum discrepancy: 4.70 percentage points, unresolved.
- No history backfill; missingness includes different start and end dates.

analysis_eda.py; outliers.csv; recomputation_check.csv

EDA implications for portfolio design



- Median annual vol: commodities 29.1%, fixed income 4.8%.
- Within-class correlation: equities 0.76, fixed income 0.62.
- Unequal histories require dynamic eligibility.
- Volatility scaling balances standalone risk; covariance still matters.

EDA outputs; additional summaries of MonthlyReturns.csv

Two baseline versions: rolling and course PPT

Dimension	Rolling baseline	PPT / adjusted baseline
Script	rolling_backtest.py	strategy_backtest_adj.py
12-month signal	sign[∏(1+r) − 1]	sign[Σr]
Volatility estimate	Trailing 12 months	Expanding; at least 36 valid months
Evaluation sample	Feb 1970–Dec 2014; 539 months	Jan 1976–Dec 2014; 468 months
TSMOM CAGR / Sharpe	19.12% / 1.06	10.64% / 0.97
TSMOM vol / MaxDD	18.05% / −27.58%	11.15% / −26.03%

Both use historical inputs, execute in the next month, and compound wealth. Different signals, risk windows, eligibility, and samples prevent a signal-only comparison.

rolling_backtest_output/performance_summary.csv: rolling_tsmom, 0 bps; strategy_backtest_adj_output/baseline_summary.csv

The course PPT approach: adjusted baseline

$$s_{i,t} = \operatorname{sign}\left(\sum_{k=011} r_{i,t-k}\right)$$

$$\hat{\sigma}_{i,t} = \operatorname{Std}(r_{i,1:t}) \times \sqrt{12}$$

$$w_{i,t} = \frac{1}{N_t} \times s_{i,t} \times \frac{0.40}{\hat{\sigma}_{i,t}}$$

- Expanding volatility requires at least 36 valid monthly observations.
- Weights formed in month t earn returns in t+1; rebalance monthly.
- Static uses the same volatility and eligibility, always long.

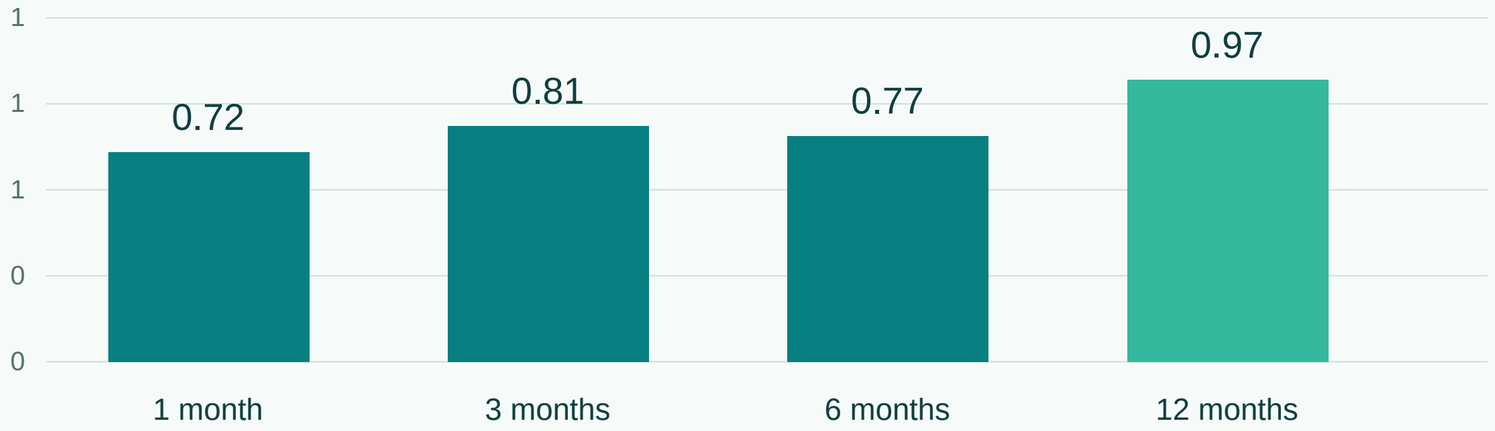
strategy_backtest_adj.py follows the course PPT approach. The 40% parameter scales individual markets; it is not a portfolio volatility target.

Execution timing and assumptions

Formation month t	Execution month t+1
Signal and risk estimates use data through t	execution_weights = weights.shift(1)
Construct equally allocated inverse-volatility positions	Rp,t□1 = Σ wi,t ri,t□1
Keep formation and execution alignment separately	Timing checks confirm a one-month lag

- Missing realized returns for held positions are filled with zero.
- No leverage cap, margin constraints, financing, or explicit roll costs.
- The diagonal risk model assumes zero cross-market covariance.

Four lookbacks: complete portfolio backtests



Lookback	1 month	3 months	6 months	12 months
CAGR	7.71%	9.23%	8.47%	10.64%
Annual vol	11.23%	11.81%	11.36%	11.15%

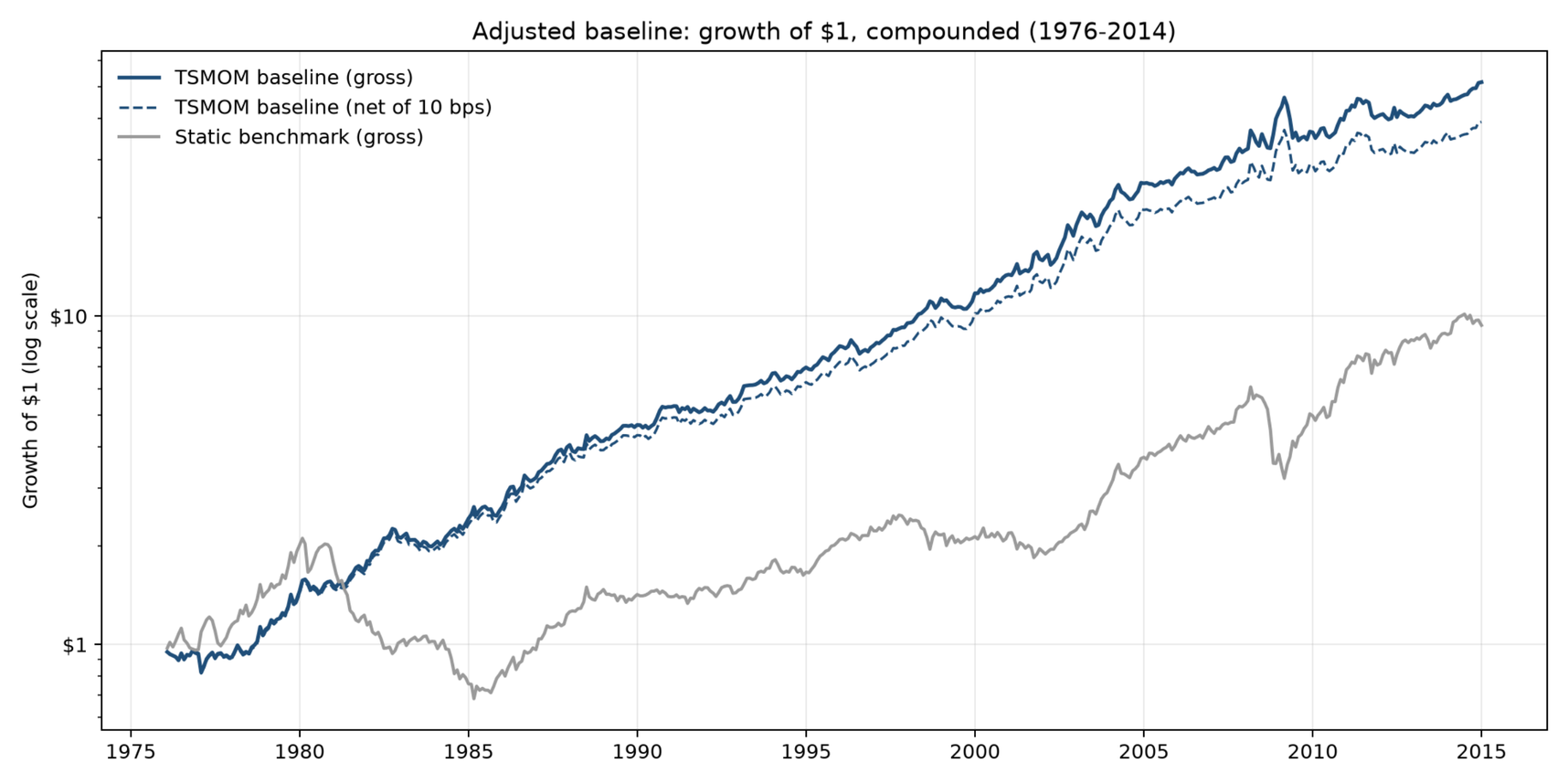
strategy_backtest_adj_output/lookback_summary.csv; 1976–2014; RF = 0

Main results: stronger returns and smaller drawdowns

1976–2014; RF = 0	PPT / adjusted TSMOM	Static long
CAGR	10.64%	5.91%
Annualized mean	10.77%	6.82%
Annualized volatility	11.15%	14.52%
Sharpe: annualized mean / vol	0.97	0.47
Maximum drawdown	−26.03%	−67.54%
Positive-return months	64.1%	60.0%
Annual turnover	7.24	0.41

baseline_performance_summary.csv□strategy_backtest_adj_output/baseline_summary.csv

Compounded wealth of the main strategy



outputs/baseline_performance_output/baseline_cumulative_return.png

Adjusted baseline: exposure and trading costs

Risk / implementation	Evidence
Gross exposure	Average 3.13; maximum 4.60
Net exposure	Average +0.84
Annual turnover	7.24 versus static 0.41
0 / 10 / 25 bps CAGR	10.64% / 9.85% / 8.68%
0 / 10 / 25 bps Sharpe	0.97 / 0.90 / 0.81

Costs are illustrative; leverage, financing, roll, and liquidity constraints remain unmodeled.

Adjusted baseline versus static: correlation

1976–2014, gross	Result
Overall TSMOM–static correlation	0.080
Class correlations: COMM / EQ / FI / FX	0.211 / −0.119 / 0.684 / −0.123
Annual mean TSMOM minus static	3.95 percentage points

Low correlation means different return behavior. It does not alone establish outperformance or crisis protection.

Rolling baseline: a second strategy specification

Component	Implementation in rolling_backtest.py
Signal	Sign of compounded trailing 12-month return
Volatility	Trailing 12 valid consecutive months × √12
Eligibility	Valid signal, valid positive volatility; matched static universe
Position	(1/N) × signal × 0.40 / volatility
Execution	Formation-month weights shifted one month
Evaluation	Feb 1970–Dec 2014; 539 months

Compounded versus summed momentum signals

Both baselines compound wealth; their historical signal aggregation differs

Example: +20%, then −18%	Calculation	Position direction
Rolling compound signal	1.20 × 0.82 − 1 = −1.6%	Short
Adjusted sum signal	20% − 18% = +2.0%	Long

- Large swings can cause opposite signal directions.
- Risk-window and eligibility differences also affect portfolio results.
- A controlled signal-only test would hold all other choices fixed.

rolling_backtest.py signal usage; strategy_backtest_adj.py: momentum_signal; illustrative example

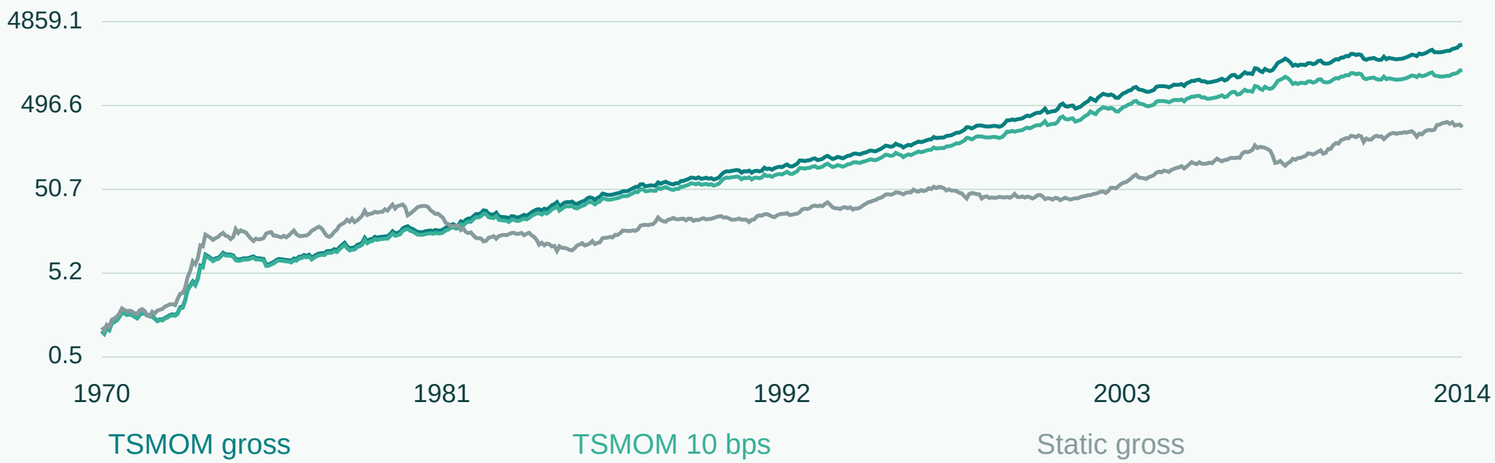
Rolling baseline: performance and cost sensitivity

Rolling baseline, Feb 1970–Dec 2014	CAGR	Annual vol	Sharpe	MaxDD
TSMOM, 0 bps	19.12%	18.05%	1.06	−27.58%
TSMOM, 5 bps	18.21%	18.05%	1.02	−27.76%
TSMOM, 10 bps	17.30%	18.06%	0.97	−27.93%
TSMOM, 25 bps	14.60%	18.12%	0.84	−28.45%
Static, 0 bps	13.33%	21.89%	0.68	−71.78%

- Compounded 12-month signal with trailing 12-month volatility.
- TSMOM and static share the same eligible universe.
- Average monthly traded notional: 1.30; compare costs separately from the PPT baseline.

rolling_backtest.py; rolling_backtest_output/performance_summary.csv: rolling_tsmom / rolling_static rows only

Rolling baseline: compounded wealth



Log wealth scale. These curves use only rolling-baseline return columns.

Rolling correlation with its static benchmark

Cost applied to both portfolios	Correlation	Annual mean TSMOM − static
0 bps	0.3846	4.29pp
5 bps	0.3839	3.77pp
10 bps	0.3831	3.26pp
25 bps	0.3805	1.71pp

- Correlation is moderate and stable across these cost scenarios.
- Positive mean differences remain, but shrink with trading costs.
- This is descriptive return difference, distinct from regression alpha.

rolling_backtest_output/portfolio_correlations.csv

Rolling regressions: benchmark and factor evidence

Regression	Gross	10 bps
Static regression annual alpha	14.43%	13.05%
Static beta / R^2	0.317 / 14.79%	0.316 / 14.68%
Six-factor annual alpha, 2000–2014	9.12%	6.38%
Six-factor alpha p-value	0.0116	0.0862

HAC standard errors use 3 lags. Regression intercepts differ from simple return differences; factor controls are incomplete.

Why do the static correlations differ?

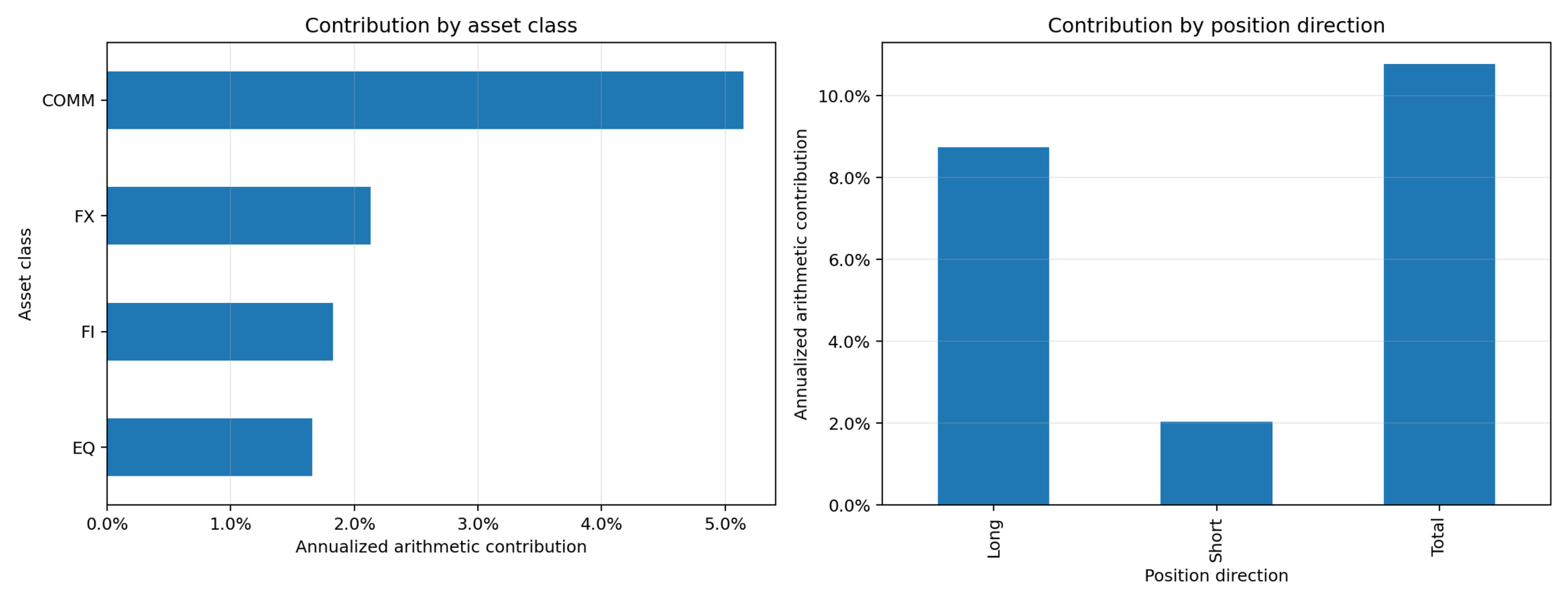
Comparison	PPT / adjusted	Rolling
Corr with own static	0.080	0.385
Signal / volatility	Sum / expanding	Compound / trailing 12m
Sample	1976–2014	1970.02–2014

- Possible mechanism: shared recent-volatility scaling may increase co-movement between rolling TSMOM and static.
- Signal signs, eligibility, and sample periods also differ; their effects are not isolated.
- Each static benchmark is different. Lower correlation alone does not imply a better strategy.

Observed gap, untested explanation. Align samples and eligibility, then change one component at a time.

Adjusted baseline_summary.csv; rolling portfolio_correlations.csv; mechanism is a hypothesis

Asset classes and trading direction generate returns



All classes and both directions contribute; commodities and longs dominate.

outputs/baseline_return_attribution_output/asset_class_and_direction_contribution.png

Return attribution by regime and decade

Internal S&P futures regime	Annualized mean	Annualized volatility
Bear	10.26%	12.60%
Bull	11.13%	8.33%

Decade	1970s	1980s	1990s	2000s	2010–2014
Annualized mean	10.37%	12.22%	9.55%	12.29%	7.57%

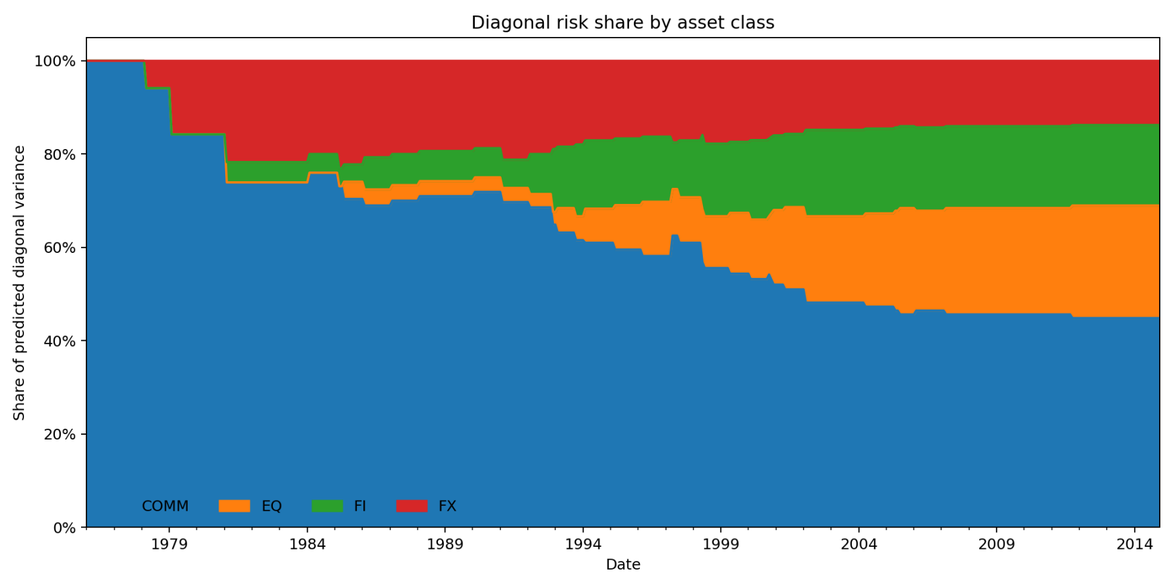
Returns are positive in every reported decade and both regimes. Bear-regime risk is higher, and recent returns are weaker.

Six-factor regression: significant momentum exposure

Joint regression, 2000–2014	Alpha / Beta	HAC t	p
Annualized alpha	7.20%	2.27	0.023
Mkt-RF	0.004	0.05	0.964
SMB	−0.005	−0.06	0.949
HML	−0.222	−1.71	0.087
Mom	0.286	5.17	<0.001
RMW	0.184	1.76	0.078
CMA	0.218	1.31	0.189

R^2 = 28.0%. Equity factors do not fully explain futures returns; residual alpha is not automatically evidence of manager skill.

Many markets, concentrated asset-class risk



- Commodities: 62.7% of average predicted variance.
- FX: 15.9%; fixed income: 11.0%; equities: 10.4%.
- Effective instrument risk positions: 39.6.
- Effective asset-class risk positions: 2.35.
- A large market count does not ensure diversified common risks.

The diagonal model understates future risk

Fixed-weight horizon	Predicted vol	Realized vol	Realized − predicted	Correlation	Realized exceeds forecast
Next 12 months	6.82%	9.78%	+2.96pp	0.13	74.8%
Next 3 months	6.79%	8.87%	+2.08pp	0.08	54.6%

- Freeze formation-date weights and estimate volatility from subsequent returns.
- At 12 months, realized risk exceeds the forecast by more than 25% in 50.4% of observations.
- Ignoring correlation among same-direction positions is a plausible source of bias.

SG Trend: shared trend exposure, different drawdowns

Common sample: 2000–2014	PPT / adjusted TSMOM	SG Trend
Annualized mean	10.72%	7.75%
CAGR	10.37%	6.90%
Annualized volatility	12.83%	14.60%
Sharpe; RF = 0	0.84	0.53
Maximum drawdown	−26.03%	−17.66%

Correlation: 0.62. Raw-return regression beta: 0.54; annualized alpha: 6.52%; HAC t: 2.48.

Four completed PPT / adjusted experiments

Experiment	Change from baseline	Purpose
E1: EWMA volatility	6-month half-life; keep 36-month eligibility	Respond faster to changing risk
E2: Asset-class budget	Equal allocation across valid classes; 1/N within class	Reduce market-count imbalance
E3: Multiple lookbacks	Average 1-, 3-, 6-, and 12-month signs	Reduce dependence on one horizon
E4: Combined	E1 + E2 + E3	Test whether improvements combine

These experiments have results. Full covariance models, portfolio volatility targeting, and walk-forward evaluation remain next-stage work.

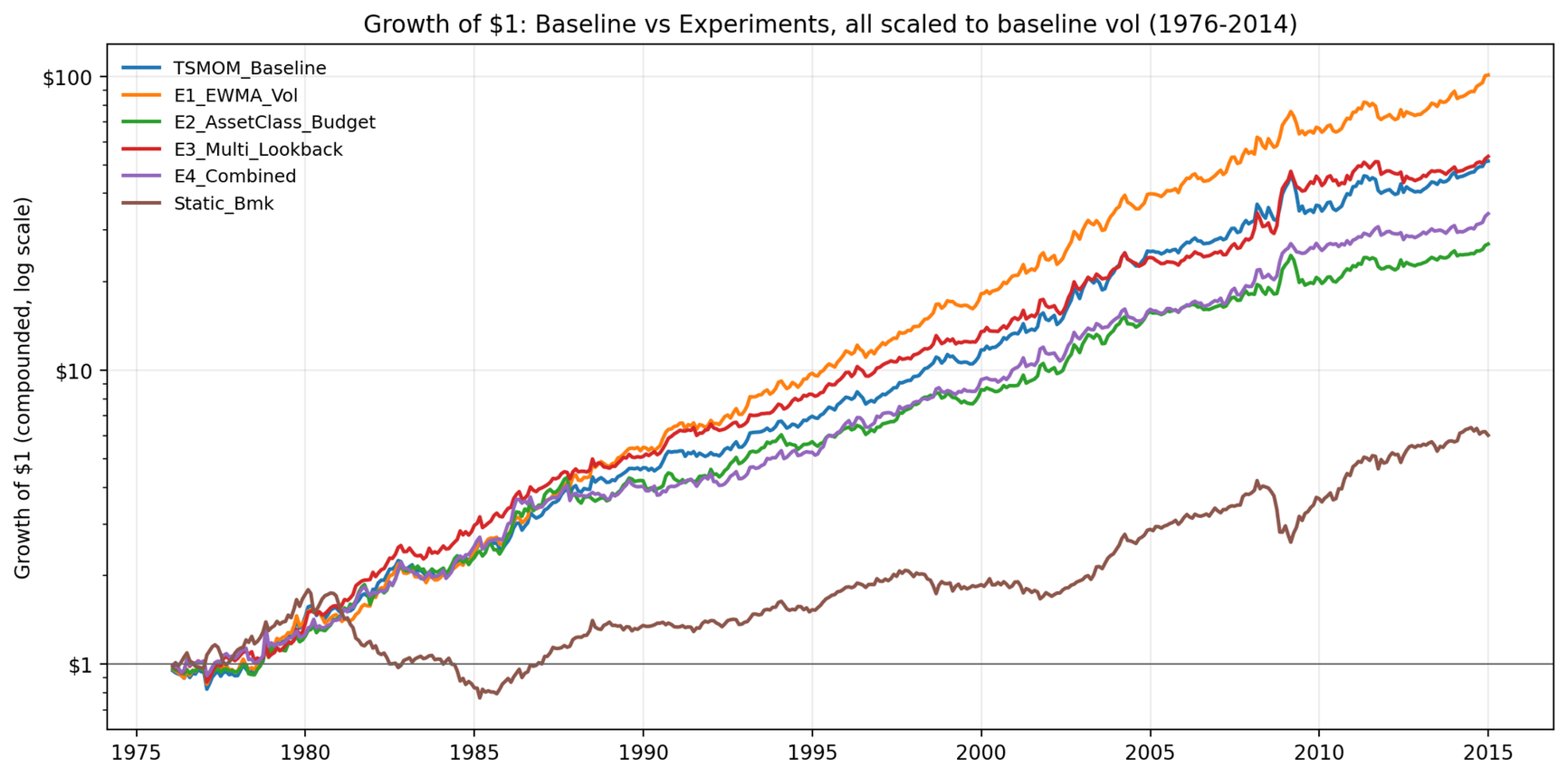
Experimental improvements do not simply add up

Strategy	Raw Sharpe	Matched-vol CAGR	Matched-vol MaxDD	Monthly turnover
Baseline	0.97	10.64%	-26.03%	0.60
E1_EWMA_Vol	1.12	12.57%	-16.63%	1.10
E2_AssetClass_Budget	0.82	8.81%	-20.81%	0.79
E3_Multi_Lookback	0.97	10.74%	-16.46%	1.31
E4_Combined	0.87	9.48%	-14.46%	2.82

- E1 reaches a 1.12 Sharpe, with substantially more turnover.
- E3 has a similar Sharpe to baseline and smaller drawdowns, with higher turnover.
- E2 and E4 have lower Sharpe; allocation and costs require further assessment.

strategy_backtest_adj_output/experiment_summary.csv

Enhancement paths at comparable volatility



Full-sample volatility matching is retrospective. All experiment results are gross.

strategy_backtest_adj_output/cumulative_experiments.png

Findings: credible signal, incomplete risk model

- Predictiveness: the 12-month signal has a 54.4% hit rate and supportive HAC evidence.
- Historical performance: PPT / adjusted CAGR of 10.64% and Sharpe of 0.97.
- Return sources: all classes and both directions contribute; commodities and longs dominate.
- Risk gap: predicted volatility of 6.82% versus subsequent realized volatility of 9.78%.
- Implementation: leverage, turnover, zero-filled returns, and omitted costs matter.
- Experiments: EWMA improves results; multiple horizons reduce drawdowns; effects are not additive.

Next steps and reproducibility

- Add covariance forecasts and validate future fixed-weight risk.
- Test portfolio volatility targeting, leverage caps, and volatility floors.
- Align baseline samples; use walk-forward tests and realistic costs.
- Resolve RL and outlier sensitivity; expand futures-relevant factors.

Reproduce rolling and adjusted baselines separately. Detailed attribution and risk validation use standardized adjusted-baseline outputs.